

Joshua Oti

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EDUCATION

University of Southern California Master of Science, Computer Science; GPA: 3.76/4.0 • Relevant Courses: Software Engineering, Database Systems, Operating Systems, Networks, Algorithms	Los Angeles, CA May 2026
Covenant University Bachelor of Engineering, Mechanical Engineering; GPA: 4.85/5.0 • Relevant Courses: CAD/CAM, Control Systems, Electrical Measurements & Instrumentation, Production Engineering • Academic Honors: Top 1% of Graduating Class (Mechanical Engineering); 3rd Place, ACPC (Regional ICPC)	Ogun, Nigeria Oct 2021

TECHNICAL SKILLS

• Languages: Python, C, C++, TypeScript, SQL, Java, C#, bash
• Hardware: Soldering, PCB Schematic Capture, Oscilloscopes, Logic Analyzers, CNC Machining
• AI/ML: PyTorch, LangChain, MCP, Transformers, LoRA, Scikit-learn, ChromaDB, RAGAS, Google AI Studio
• Tools: Git, RabbitMQ, Kafka, Figma, GraphQL, Grafana, Prometheus, Postman, JMeter, MLflow
• Cloud/Infra: Microsoft Azure, Amazon Web Services, Google Cloud, Docker, CI/CD
• Certifications: AWS Certified Solutions Architect Associate, CSWP-Mechanical Design, CSWA-Additive Manufacturing, Autodesk Certified User (AutoCAD, Inventor)
• Web/Backend: FastAPI, gRPC, React, Vue.js, Flask, Node.js, Spring Boot, ASP.NET Core, Django, OAuth 2.0

EXPERIENCE

Autodesk Machine Learning Engineer Intern • Owned the end-to-end lifecycle of an enterprise GenAI content platform using a FastAPI backend, Azure OpenAI, and AWS S3, automating workflows to save 10+ hours weekly. • Designed dynamic LLM pipelines incorporating few-shot prompting strategies to significantly optimize generative model reasoning and image output quality. • Engineered a responsive Vue.js gamified frontend with enterprise SSO integration, driving a 20% increase in user participation through seamless, multi-device accessibility.	San Francisco, CA May 2025 - Aug 2025
Phasecurve Technologies Ltd Software Engineer • Architected a cloud-based ETL and analytics backend using Azure SQL and Flask microservices on Azure App Services, implementing autoscaling to absorb sudden 3x traffic spikes on a fintech platform. • Engineered event-driven data pipelines using Azure Serverless Functions and Azure Queues for asynchronous social media data extraction, implementing exponential backoff to reliably handle strict third-party API rate limits. • Automated CI/CD pipelines leveraging GitHub Actions and Docker, containerizing services to ensure reliable deployments and cutting deployment time by 50%. • Optimized content uploads with multi-threading and asynchronous processing, reducing backend response latency by 80%, and integrated Azure Application Insights for telemetry monitoring to rapidly diagnose pipeline failures. • Led the migration from Mailgun API to Gmail API, optimizing the email routing service and significantly reducing operational costs.	Lagos, Nigeria Nov 2022 - Sep 2024
ReHboX Machine Learning Engineer • Developed and deployed a real-time computer vision algorithm for edge devices to analyze human pose from camera inputs utilizing Python and OpenCV. • Built a predictive machine learning model using Scikit-learn alongside an interactive data dashboard, providing actionable insights on the Total Addressable Market (TAM). • Coauthored 6 technical design documents, aligning engineering workflows with user stories to streamline development sprints and cut meeting times by 15%.	Lagos, Nigeria Jun 2021 - Feb 2022

SELECTED PROJECTS

Distributed Healthcare Network C++, TCP/UDP, SHA-256 — Systems & Security • Architected a multi-tier distributed network comprising a client UI, central router, and three scalable backend services to optimize high-throughput data routing. • Managed concurrent TCP client connections and routed backend queries via UDP sockets, securing sensitive PII/PHI with SHA-256 cryptographic hashing for HIPAA compliance. • Programmed low-level Unix socket routines and custom signal handlers to automate child process reaping, preventing memory leaks and ensuring fault-tolerant socket teardowns.	Jan 2026
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Concurrent Request Orchestrator | C, React — Operating Systems

Sep 2025

- Engineered a multithreaded resource orchestrator in C with a React frontend to visualize and optimize OS-level task scheduling and concurrency control.
- Programmed POSIX threads (pthreads), mutexes, and condition variables to manage thread synchronization, preventing race conditions and guaranteeing atomic access to shared job queues.
- Developed a real-time React dashboard to monitor system throughput, buffer occupancy, and thread activity, enabling the rapid identification and resolution of pipeline bottlenecks.

DevPortalX | React, Spring Boot, OAuth 2.0, DynamoDB — Full-Stack & Security

Oct 2024

- Engineered an enterprise developer portal and centralized Identity Provider (IdP) using React and Spring Boot to manage third-party API integrations.
- Architected a custom OAuth 2.0 authorization server utilizing Spring Security to enable Single Sign-On (SSO) for external client applications.
- Implemented role-based access controls (RBAC) by issuing and validating JWTs to secure API endpoints and user data.
- Integrated DynamoDB for highly scalable data storage alongside analytics tooling, providing end-users with actionable API usage metrics and streamlined API key provisioning.

LEADERSHIP

• National Society of Black Engineers, Graduate Student Liaison USC Chapter

Apr 2025 - Apr 2026

Bridged the undergraduate-graduate gap by organizing joint social initiatives, driving a 25% increase in graduate student event attendance.